

# Nicole Sandra-Yaffa Dumont

📍 Zürich, Switzerland

✉ Email: [nidumon@ini.uzh.ch](mailto:nidumon@ini.uzh.ch)

🌐 Website: [nsdumont.github.io](https://nsdumont.github.io)

in LinkedIn: [nicole-dumont](#)

🐙 Github: [nsdumont](#)

🔗 Scholar: [2hnh9gkAAAAJ](#)

🆔 ORCID: [0000-0001-5041-1527](#)

## Academic Appointments

### Postdoctoral Researcher

Institute of Neuroinformatics (University of Zurich & ETH Zurich)

📅 Dec 2025 – Present

📍 Zürich, Switzerland

- **Supervisor:** Prof. Katharina Wilmes
- Awarded the **UZH Postdoc Grant**, beginning December 2026.
- **Predictive Coding (PC) & Models of Perception and Learning in Cortex**
  - Comparing prediction error modulation by learned uncertainty signals and how this mechanism differs from top-down, goal-conditioned attentional modulation.
  - Developing PC networks for multi-sensory integration to better understand integration circuits in the brain.

## Education

### Ph.D in Computer Science

Diploma in Theoretical Neuroscience

University of Waterloo

📅 Sept 2019 – April 2025

📍 Waterloo, Canada

- **Supervisors:** Prof. Chris Eliasmith and Prof. Jeff Orchard
- **Dissertation:** *Symbols, Dynamics, and Maps: A Neurosymbolic Approach to Spatial Cognition* [\[Link\]](#)
- **Neuro-Symbolic Modeling for Spatial Representation & Mapping**
  - Developed Spatial Semantic Pointers (SSPs) as a grid cell-inspired representation of continuous features, enabling the integration of symbolic structure and uncertainty in biologically plausible neural models.
  - Built spiking neural network models of path integration, cognitive mapping, and SLAM that use neurosymbolic methods to bind continuous and discrete features to create compositional embeddings.
- **Reinforcement Learning (RL) & Action Selection**
  - Designed biologically plausible online learning rules for RL in continuous-time settings using State Space Model-based memory systems.
  - Incorporated SSPs into Bayesian Optimization and deep RL frameworks, modeling predictive inference and decision-making under uncertainty.

### Computational Mathematics (Masters of Mathematics – Co-op Program)

University of Waterloo

📅 Sept 2017 – April 2019

📍 Waterloo, Canada

### Honours Mathematics and Physics (Bachelors of Science)

McMaster University

📅 Sept 2012 – April 2017

📍 Hamilton, Canada

## Work Experience

### Research Associate

Computational Neuroscience Research Group, University of Waterloo

📅 April 2025 – Nov 2025

📍 Waterloo, Canada

- Contributed to the incorporation of spatial reasoning and embodiment in Spaun 3.0, a large-scale functional brain model.

### Neuromorphic Scientist

Applied Brain Research

📅 Jan 2023 – Jan 2024

📍 Waterloo, Canada

- Integrated a biologically-inspired SLAM algorithm into a MuJoCo simulation environment and deployed it on low-power neuromorphic hardware (Intel Loihi chip).

## Research Associate

Cayuga Research

📅 May 2018 – Aug 2019

📍 Waterloo, Canada

- Consulting work focused on developing optimization methods and data-driven solutions for industrial applications.

## Research Assistant

Ayers Research Group, McMaster University

📅 May 2015 – Aug 2015

📍 Hamilton, Canada

## Research

h-index = 7, i10-index = 5, total citations = 171 (Google Scholar on 25/08/2026)

† = These authors contributed equally

### Publications & Conference Proceedings

- Furlong, P.M.<sup>†</sup>, **Dumont, N.S.**<sup>†</sup>, Antonova, R.<sup>†</sup>, Orchard, J., & Eliasmith, C. (2026). Compositional neurosymbolic representations enable efficient active exploration. *Nature Communications*. <https://doi.org/10.1038/s41467-026-75703-4> [Link]
- **Dumont, N.S.**, Furlong, P. M., Orchard, J., & Eliasmith, C. (2023). Exploiting semantic information in a spiking neural slam system. *Frontiers in Neuroscience*, 17 [Link]
- **Dumont, N.S.**<sup>†</sup>, Stöckel, A.<sup>†</sup>, Furlong, P. M.<sup>†</sup>, Bartlett, M., Eliasmith, C., & Stewart, T. C. (2023). Biologically-based computation: How neural details and dynamics are suited for implementing a variety of algorithms. *Brain Sciences*, 13(2) [Link]
- Voelker, A. R., Blouw, P., Choo, X., **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Simulating and predicting dynamical systems with spatial semantic pointers. *Neural Computation*, 33(8) [Link]
- Coleman, T. F., **Dumont, N.S.**, Li, W., Liu, W., & Rubtsov, A. (2021). Optimal pricing of climate risk. *Computational Economics* [Link]

### Conference Proceedings & Posters

- **Dumont, N.S.** & Wilmes, K. A. (2026). Multi-sensory integration in predictive processing [Accepted, poster presentation scheduled for Bernstein Conference, Sep 28 – Oct 1, 2026]
- Simone, K., Steffen, L. C., **Dumont, N.S.**, Yu, S., Ly, H., Damberger, G., & Eliasmith, C. (2026). Semantic map learning and externalization in an embodied neural agent: A comparison to human behavioral and neural data. *CogSci*, 48(48)[Link]
- Furlong, P. M., Simone, K., **Dumont, N.S.**, Bartlett, M., Stewart, T. C., Orchard, J., & Eliasmith, C. (2024). Biologically-plausible markov chain monte carlo sampling from vector symbolic algebra-encoded distributions. *ICANN* [Link]
- Furlong, P. M., **Dumont, N.S.**, & Orchard, J. (2024). A recurrent dynamic model for efficient bayesian optimization. *NICE* [Link]
- Bartlett, M., Simone, K., **Dumont, N.S.**, Eliasmith, C., & Orchard, J. (2023). Improving reinforcement learning with biologically motivated continuous state representations. *ICCM* [Link]
- **Dumont, N.S.**, Orchard, J., & Eliasmith, C. (2022). A model of path integration that connects neural and symbolic representation. *CogSci*, 44(44) [Link]
- Bartlett, M.<sup>†</sup>, **Dumont, N.S.**<sup>†</sup>, Furlong, M. P., & Stewart, T. C. (2022). Biologically-plausible memory for continuous time reinforcement learning. *ICCM* [Link]
- **Dumont, N.S.**, Stewart, T. C., & Eliasmith, C. (2021). Spiking neural network model of simultaneous localization and mapping with spatial semantic pointers [Poster presented at CoSyNe, February 23–26, 2021] [Link]
- **Dumont, N.S.** & Eliasmith, C. (2020). Accurate representation for spatial cognition using grid cells. *CogSci* [Link]

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## Manuscripts

- Sainz Villalba, L., Furlong, P. M., Bartlett, M., & **Dumont, N.S.** (2026). Neural representational geometry of feature binding operations. *bioRxiv* [\[Link\]](#)

## Patents

- Voelker, A. R., Eliasmith, C. D., Blouw, P., Stewart, T., & **Dumont, N.S.** (pending, filed November 5, 2021). Methods and systems for simulating and predicting dynamical systems with vector-symbolic representations [US Patent App. 17/520,379] [\[Link\]](#)

## Teaching and Supervision

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- **Masters supervision (2026-present):** Supervising a masters student thesis project on learning compositional neural representation for planning and reasoning with tree diffusion.
- **Student supervision (2020):** Supervised an undergraduate research assistant during a four-month project developing a novel algorithm for learning successor features for reinforcement learning.
- **Introduction to Computational Mathematics (2022):** Sole instructor for two sections of this course, teaching over 120 students. Planned course content and developed lectures, assignments, and exams.
- **Logic and Computation (2020):** Served as an Instructional Apprentice for an introductory course in mathematical logic. Led tutorials, graded assignments, and provided assistance during office hours.
- **Teaching Assistantships (2017-2022):** Served as a TA in six undergraduate courses in mathematics and computer science.

## Academic Service

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- **Leadership & Organization**
  - Topic Area Leader, Spatiotemporal Dynamics in Neural Computation, Telluride Neuromorphic AI Workshop (2026)
  - Topic Area Leader, Language and Thought, Telluride Neuromorphic Cognition Engineering Workshop (2025): Successfully proposed and led a topic area; delivered tutorials on cognitive mapping models; mentored PhD-level researchers through the full project lifecycle from conceptualization to demonstrations for the public.
  - Invited Speaker & Co-Organizer, Telluride Neuromorphic Cognition Engineering Workshop (2024)
- **Instruction & Training**
  - Instructor, Nengo Summer School (2023, 2024, 2025): Led tutorials and mentored participants developing neural models using the Nengo framework.
- **Peer Review:** Reviewer for *Nature Communications*, *Communications Biology*, *CogSci 2025*, and *Frontiers in Computational Neuroscience*

## Awards & Distinctions

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- UZH Postdoc Grant (2026)
- Marie Skłodowska-Curie Postdoctoral Fellowship (MSCA-PF) Seal of Excellence (2026)
- Best New Neuromorph, Telluride Neuromorphic Cognition Engineering Workshop (2024)
- European Neural Network Society Best Paper Award (2024)
- Barbara Hayes-Roth Award for Women in Math and Computer Science (2022)
- GO-Bell Scholarship (2019–2021)
- Provost Doctoral Entrance Award for Women (2019)
- Keith & Debbie Geddes Graduate Scholarship (2019)
- University of Waterloo Graduate Scholarship (2018–2019)
- Emanuel Williams Scholarship in Physics (2014)